Android Debug Bridge (ADB) and Mobile device connection with Android Studio

By

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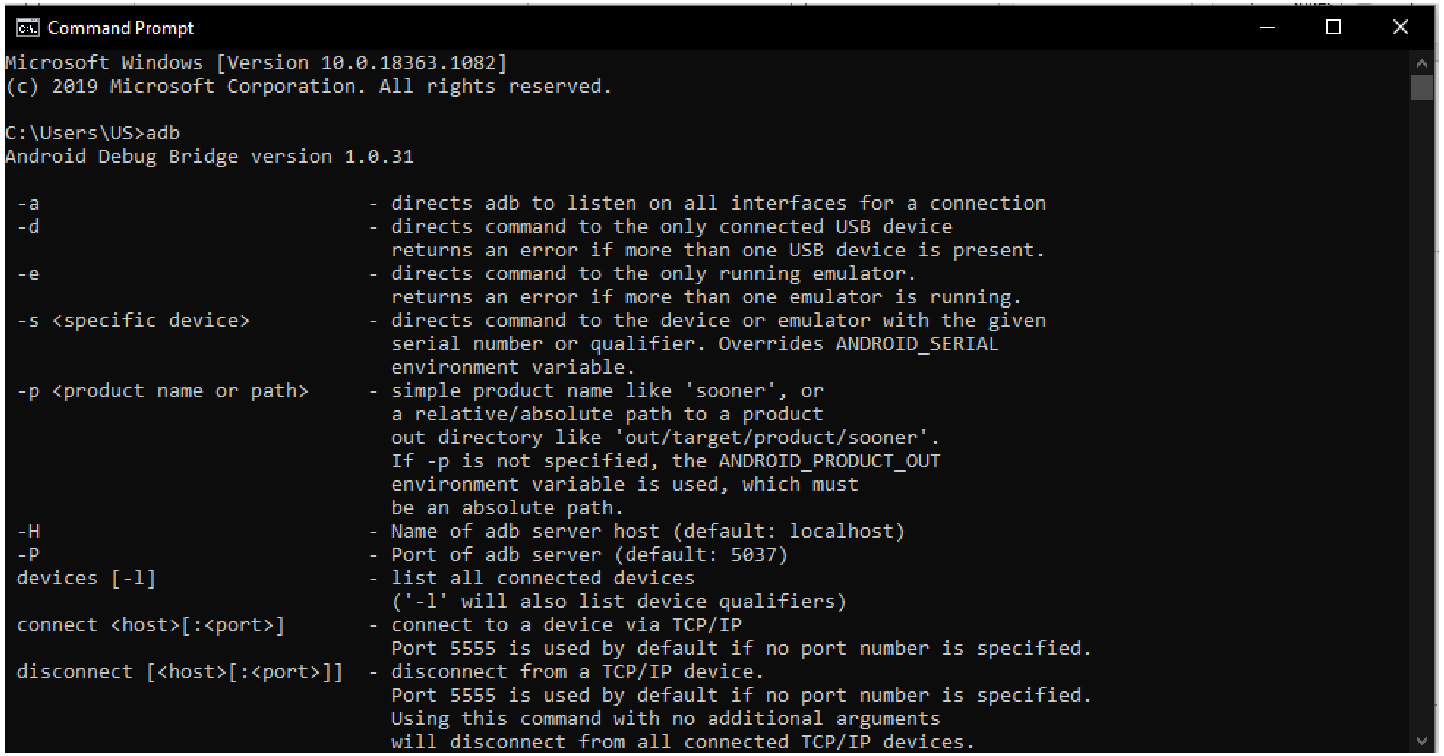
ADB Introduction & Installation

- Android Debug Bridge (adb) is a versatile command-line tool that lets you communicate with a device.
- The adb command facilitates a variety of device actions, such as installing and debugging apps, and it provides access to a Unix shell that you can use to run a variety of commands on a device
- Download universal ADB drivers
    - https://adb.clockworkmod.com/

ADB verification

- Open command prompt and type “adb”
- Check if your system detect the adb drivers



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Command Prompt
Microsoft Windows [Version 10.0.18363.1082]
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C:\Users\US>adb
Android Debug Bridge version 1.0.31

 -a                            - directs adb to listen on all interfaces for a connection
 -d                            - directs command to the only connected USB device
                                 returns an error if more than one USB device is present.
 -e                            - directs command to the only running emulator.
                                 returns an error if more than one emulator is running.
 -s <specific device>          - directs command to the device or emulator with the given
                                 serial number or qualifier. Overrides ANDROID_SERIAL
                                 environment variable.
 -p <product name or path>     - simple product name like 'sooner', or
                                 a relative/absolute path to a product
                                 out directory like 'out/target/product/sooner'.
                                 If -p is not specified, the ANDROID_PRODUCT_OUT
                                 environment variable is used, which must
                                 be an absolute path.
 -H                            - Name of adb server host (default: localhost)
 -P                            - Port of adb server (default: 5037)
 devices [-l]                  - list all connected devices
                                 ('-l' will also list device qualifiers)
 connect <host>[:<port>]       - connect to a device via TCP/IP
                                 Port 5555 is used by default if no port number is specified.
 disconnect [<host>[:<port>]]  - disconnect from a TCP/IP device.
                                 Port 5555 is used by default if no port number is specified.
                                 Using this command with no additional arguments
                                 will disconnect from all connected TCP/IP devices.
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Enabling developer mode

- Go to your mobile setting and search for build number use the following (the location might be different for different mobile models and brands)
- Setting>About>Software Info>Build number
- Click on build number for 5-6 times until the message appears the your developer mode is enabled (some device might required your screen lock to enable developer mode)

Enable USB debugging in Developer mode

- Enable developer option in developer mode
- The search for USB debugging and enable it
- Now connect your android device and accept the request to connect your device with ADB drivers from your mobile screen
- To confirm your mobile device connection open command prompt and type “adb devices”
- If it shows your devices then you are connection was successful, other wise try those commands
- abd kill-server
- adb start-server